

ER: Requirements Specification Component

With the growth of the Internet, misinformation has become and continues to be a global problem, often leading people to make poor decisions due to a lack of factual knowledge about events. Our goal is to create a platform where users can share news, while fact-checkers and administrators ensure that content is properly sourced, and where discussion is fostered among people from diverse backgrounds, perspectives and beliefs.

Our mission is to promote No-Mis-Information and No-Miss-Information.

A1: NoMisHub

The NoMisHub is being developed by a small team targeted at users who want to have quick and easy access to updated news while, simultaneously, interacting with a wider community sharing their thoughts and experiences on events influencing national and international lives.

The main goal of this project is the development of a web-based platform in which users can have access, on their home or on the go, to reliable information, curated by a trusted team of moderators and administrators. At the same time, the platform aims to promote discussion among users about diverse themes from around the world.

To make this a reality, this platform will allow registered users to post their own news and respective sources, to react and comment news and discuss with other users on a certain news piece. Users are organized into groups with different permissions. These include administrators, who have global responsibilities for user and platform management; moderators (fact-checkers), who validate news, verify sources, and apply content policies; registered users, who can interact and contribute content; and non-authenticated users, who may explore publicly available content without creating an account, though they will not be able to interact with it.

The platform will have an adaptive design, ensuring a pleasant browsing experience across devices (desktop, tablet, or smartphone). It will provide filtering and advanced search features, such as browsing by theme or most recent items. It will also provide a modern interface and a consistent, high-quality user experience. In addition, the system will be thoroughly documented to support both users and developers.

A2: Actors and User stories

This artifact identifies the main types of users (actors) who will interact with the NoMisHub and specifies the different ways in which they can use the platform. Each actor is defined with a clear role and set of responsibilities, reflecting the permissions and features available to them.

The goal of this artifact is to ensure that all user groups are clearly identified, their responsibilities understood, and their relation to the system’s functionalities established.

1. Actors

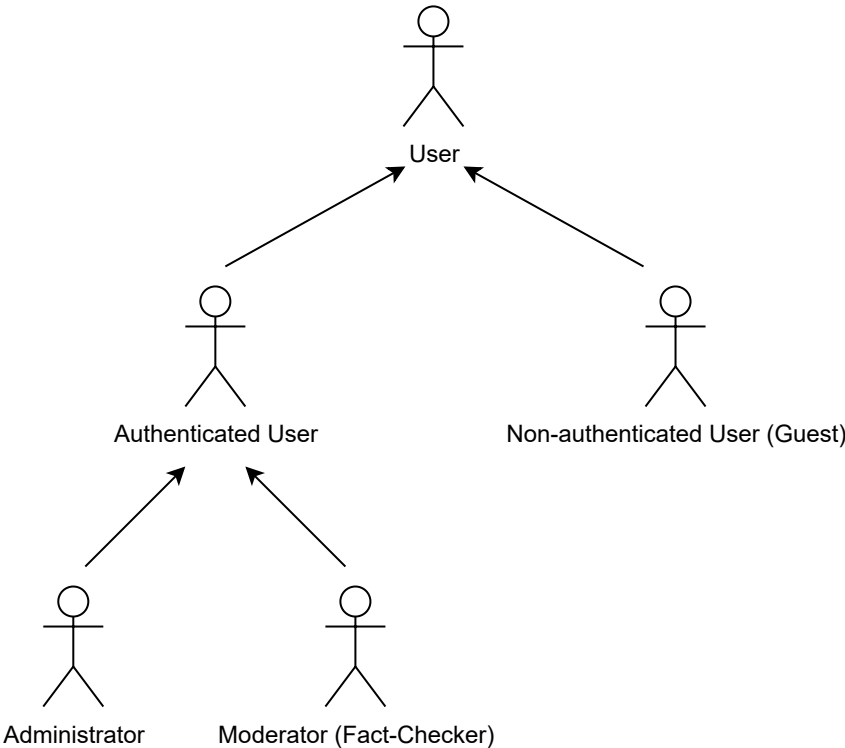


Figure 1: NoMisHub Actors Diagram and their relationships.

For the NoMisHub system, the actors are represented in Figure 1 and described in Table 1.

Actor	Description
User	Any user can read news, search for content, read comments and reactions, and consult the community rules.

Actor	Description
Non-Authenticated User (Guest)	Visitor of the platform who can view public content and see the options to log in or register.
Authenticated User	Registered user who can comment, react, publish news, manage their profile, and report content (reports generate notifications for administrators).
Moderator (Fact-Checker)	Authenticated User invited by an administrator who accepted the role of moderator. Responsible for verifying the sources of news, validating posts, publishing the results of verification processes, and applying strikes to accounts either based on reports or at their own judgment. All strikes are later reviewed by the administrator. In the user interface, moderators are displayed as <i>Fact-Checkers</i> .
Administrator	Authenticated User that is responsible for the management of Users (create/edit/remove accounts) and for some specific supervisory functions, like configure policies and remove content. They have high permissions.

Table 1: NoMisHub actors and descriptions.

2. User Stories

2.1. User

Identifier	Name	Priority	Responsible	Description
US01	Top News	Prototype	Tomás Silva	As a user, I want to be able to see which are the current top news, so that I can be on par with the most important news of the time.
US02	News Item	Prototype	Tomás Silva	As a user, I want to be able to see more information about a news item and its comments, so that I can interact with it and get to know more about it.
US03	Exact Match Search	Prototype	Tomás Silva	As a user, I want to have access to be able to perform an exact match search, so that I can find a news article or comment by its title/content.
US04	Full-text Search	Prototype	Tomás Silva	As a user, I want to have access to full-text search, so that I can find the news piece or comment I'm searching for without knowing its title/content.
US05	Recent News	Product	Tomás Silva	As a user, I want to be able to see which are the most recent news, so that I can be on par with the current happenings.
US06	Comments	Product	Tomás Silva	As a user, I want to be able to view news' comments, so that I can know about others' opinions on the matter.
US07	Search over Multiple Attributes	Product	Tomás Silva	As a user, I want to be able to perform searches over multiple attributes, so that I can easily narrow down the results of my searches.
US08	Search Filters	Product	Tomás Silva	As a user, I want to be able to perform apply search filters while searching, so that I can easily narrow down my results.
US09	Form Help	Product	Ana Patrício	As a user, I need to have information on how to fill out forms, so that I can properly fill them with minimal problems.
US10	Error Messages	Product	Ana Patrício	As a user, I need to have proper error messages that indicate what I'm doing wrong, so that I can correct said errors.

Identifier	Name	Priority	Responsible	Description
US11	About Us	Product	António Pais	As a user, want to have access to an "About Us" page, so that I can get to know important information about the website.
US12	Main Features	Product	António Pais	As a user, I want to have a page that contains the website's main features, so that I know what functionalities the website has available for me.
US13	Contacts	Product	António Pais	As a user, I need to have access to the website's managers, so that I can contact them in case of need.
US14	See Checkmark	Innovation	Tomás Silva	As a user, I want to see whether a news piece has been fact-checked by the moderators, so that I know whether a news piece is reliable or not.

Table 2: User user stories.

2.2. Non-Authenticated User (Guest)

Identifier	Name	Priority	Responsible	Description
US15	Login	Prototype	Ana Patrício	As a guest, I want to be able to log in into my account, so that I can have access to functionalities related to my account.
US16	Registration	Prototype	Ana Patrício	As a guest, I want to be able to register a new account, so that I can create an account and get access to more functionalities.
US17	Recover Password	Product	Ana Patrício	As a guest, I want to be able to recover my password, so that, in case I forget it, I can recover access to my account.

Table 3: Guest user stories.

2.3. Authenticated User

Identifier	Name	Priority	Responsible	Description
US18	Logout	Prototype	Ana Patrício	As an authenticated user, I want to be able to log out my account, so that I can end my session.
US19	View Profile	Prototype	Ana Patrício	As an authenticated user, I want to be able to view my profile, so that I can check my information.
US20	Manage Profile	Prototype	Ana Patrício	As an authenticated user, I want to be able to edit my profile, so that I can update of my personal information.
US21	User News Feed	Prototype	António Pais	As an authenticated user, I want to be able to view a personalized news feed based on my follows, so that I can quickly find news that are more relevant for my interests.
US22	News Item Details	Prototype	António Pais	As an authenticated user, I want to be able to see details about a particular news item, so that I can quickly conclude whether it's recent news, how appreciated it is among others.

Identifier	Name	Priority	Responsible	Description
US23	Publish News Item	Prototype	António Pais	As an authenticated user, I want to be able to publish a news item, so that I can share any news I have with other users.
US24	View Others' Profiles	Prototype	Ana Patrício	As an authenticated user, I want to be able to see other users' profiles so that I can see more information about them like reputation and posted articles and comments.
US25	Manage News Item	Prototype	António Pais	As an authenticated user, I want to be able to edit or delete news items that I posted, so that I can update them with accurate information or delete them needed.
US26	Vote on Content	Product	António Pais	As an authenticated user, I want to be able to vote on published content like news items and comments, so that I can easily express my view on it.
US27	Comment on News Item	Product	António Pais	As an authenticated user, I want to be able to comment on a news item, so that I can express complex opinions on it.
US28	Delete Account	Product	Ana Patrício	As an authenticated user, I want to be able to delete my account, so that my account can be terminated whenever I don't need/want it any more.
US29	Manage Profile Picture	Product	Ana Patrício	As an authenticated user, I want to be able to add or update my profile picture, so that other users can more easily identify my account.
US30	Notifications	Product	António Pais	As an authenticated user, I want to be able to receive and check my notifications, so that I know of important things that have happened on my account or content like likes and comments.
US31	Mark Notification as Read	Product	António Pais	As an authenticated user, I want to be able to mark a notification as read, whenever I have acknowledged it, so that it won't clutter my notification list.
US32	Comment Details	Product	Tomás Silva	As an authenticated user, I want to be able to see details about a particular comment, so that I can quickly conclude how recent and appreciated it is, among other information.
US33	Reputation	Product	Ana Patrício	As an authenticated user, I want to be able to see other users' reputations, so that I can have an idea of how reliable they are.
US34	View News Tags	Product	Tomás Silva	As an authenticated user, I want to be able to see a news item tags, so that I can, before reading it, whether it refers to themes that I'm interested in or not.

Identifier	Name	Priority	Responsible	Description
US35	Follow/Unfollow Users and Tags	Product	Hugo Alves	As an authenticated user, I want to be able to see follow/unfollow specific users and tags, so that I can get notified of and recommended content that might interest me.
US36	Suggest Tags	Product	Hugo Alves	As an authenticated user, I want to be able to suggest new tags to the administrators, so that they may add it to allow greater diversity and accuracy on news classification.
US37	Manage Comments	Product	Hugo Alves	As an authenticated user, I want to be able to edit or delete my own comments, so that I can update or delete them when needed.
US38	Report User or Content	Product	Hugo Alves	As an authenticated user, I want to be able to report a user or content that may violate community rules to the administrators, so that they are informed and can apply proper punishment if fit.
US39	Respond to Promotion	Innovation	Hugo Alves	As an authenticated user, I want to be able to answer to invitations sent by an administrator to become an administrator or moderator.

Table 4: Authenticated User user stories.

2.4. Administrator

Identifier	Name	Priority	Responsible	Description
US40	Manage User Accounts	Prototype	Hugo Alves	As an administrator, I want to have tools that would allow me to manage other users' accounts like editing, banning and deleting, so that I can take appropriate action if a user infringes the rules.
US41	Manage User Content	Product	Hugo Alves	As an administrator, I want to have tools that would allow me to manage other users' content like editing or deleting, so that I can take appropriate action if a certain piece of content infringes the rules.
US42	Manage Tags	Product	Hugo Alves	As an administrator, I want to have access to tools to allow the management of content tags, like receiving user suggestions, adding new tags, editing or removing them.
US43	Respond to Reports	Product	Hugo Alves	As an administrator, I want to be able to receive user reports of content or users that violate the rules.
US44	Promote to Administrator	Innovation	Hugo Alves	As an administrator, I want to be able to invite a user to become an administrator, so that the administration team can be expanded.
US45	Promote to Moderator	Innovation	Hugo Alves	As an administrator, I want to be able to invite a user to become a moderator, so that they can gain access to moderator exclusive abilities.

Table 5: Administrator user stories.

2.5. Moderator

Identifier	Name	Priority	Responsible	Description
US46	Add Checkmark	Innovation	Tomás Silva	As a moderator, I want to be able to mark a certain news piece as trustworthy after verifying it so that regular users are aware whether a news item has been fact-checked or not.
US47	Time-out Users	Innovation	Hugo Alves	As a moderator, I want tools to temporarily restrict a user's account (time-out), so that I can take appropriate action when a user infringes the rules without imposing a permanent ban, while automatically notifying the admins of the action.

Table 6: Moderator user stories.

3. Supplementary Requirements

3.1. Business rules

Identifier	Name	Description
BR01	Account Deletion	Upon account deletion, public user data, like comments, news articles and likes are not erased, but anonymised
BR02	Reputation Value	Each user's reputation is based on the likes and dislikes they received on their published content (comments, news articles, etc.)
BR03	Content Deletion	A user cannot remove their published content if it has received any votes or comments
BR04	Checkmark	A moderator can't mark their own news article as trustworthy (with the checkmark) to maintain help maintain the position's integrity in the eyes of normal users.
BR05	User Votes	Users can vote on their own posted content.
BR06	Comment vs News Dates	The publishing date of posting of a comment to a news article must be greater than the date of posting of the respective news article.

Table 7: NoMisHub business rules.

3.2. Technical requirements

Identifier	Name	Description
TR01	Performance	The system should have response times shorter than 2s in order to keep the user's attention.
TR02	Robustness	The system must be prepared to handle and continue operating when runtime errors occur, so that users are not disincentivised of searching for the information they need because of said errors, ensuring they have access to the information.
TR03	Scalability	The system must be prepared to deal with the growth in the number of users and their actions. NoMisHub is projected to receive all kinds of news from all around the world, as such, it is very important and expectable that the system is capable of handling vast amounts of information.
TR04	Accessibility	The system must ensure that everyone can access all the content, regardless of any handicaps they may have or the Web browser they use. Information should be freely and readily available to everyone, regardless of any limitations they may have or from where they access it. As such, NoMisHub must support accessibility features destined to the needs of users with disabilities and support all browsers compliant with the W3C standards.

Identifier	Name	Description
TR05	Security	The system must store user passwords securely. One of our main objectives is the distribution of reliable news. This reliability is to be maintained by our team of moderators, as such, it is of utmost importance that the system is protected against unauthorized access and data breaches.

Table 8: NoMisHub technical requirements.

3.3. Restrictions

Identifier	Name	Description
R01	Deadline	The system should be ready to be used around the middle of December, before the holidays, to be able to get the first news of the following year.

Table 9: NoMisHub project restrictions.

A3: Information Architecture

This artifact has as its main purpose to show the way the information will be presented on our platform via a sitemap and wireframes.

1. Sitemap

The image below represents a simple design of how the pages in NoMisHub can be accessed or used.

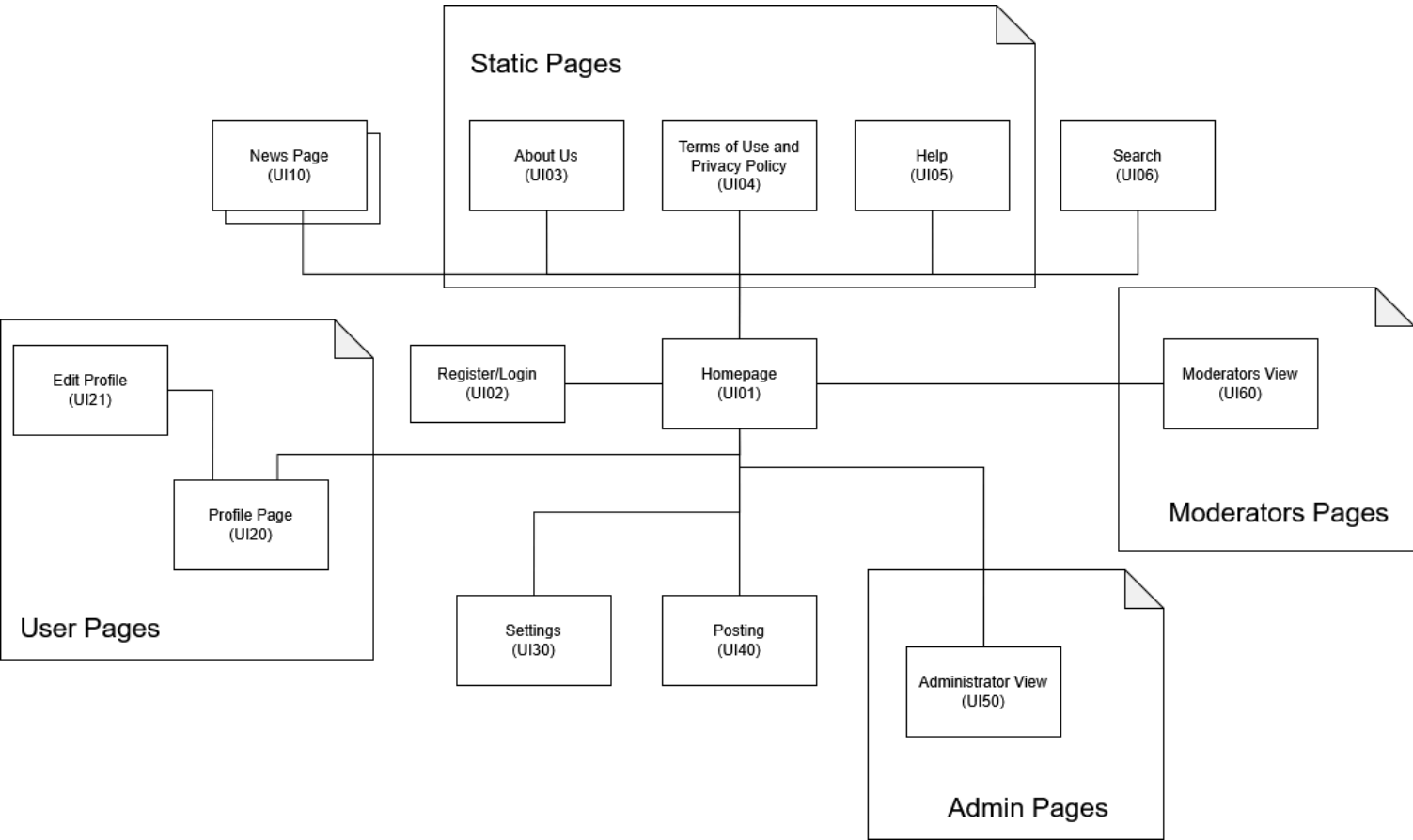


Figure 2: Sitemap.

2. Wireframes

The wireframes below show the layout and position of the main interactive elements.

UI01: Home Page

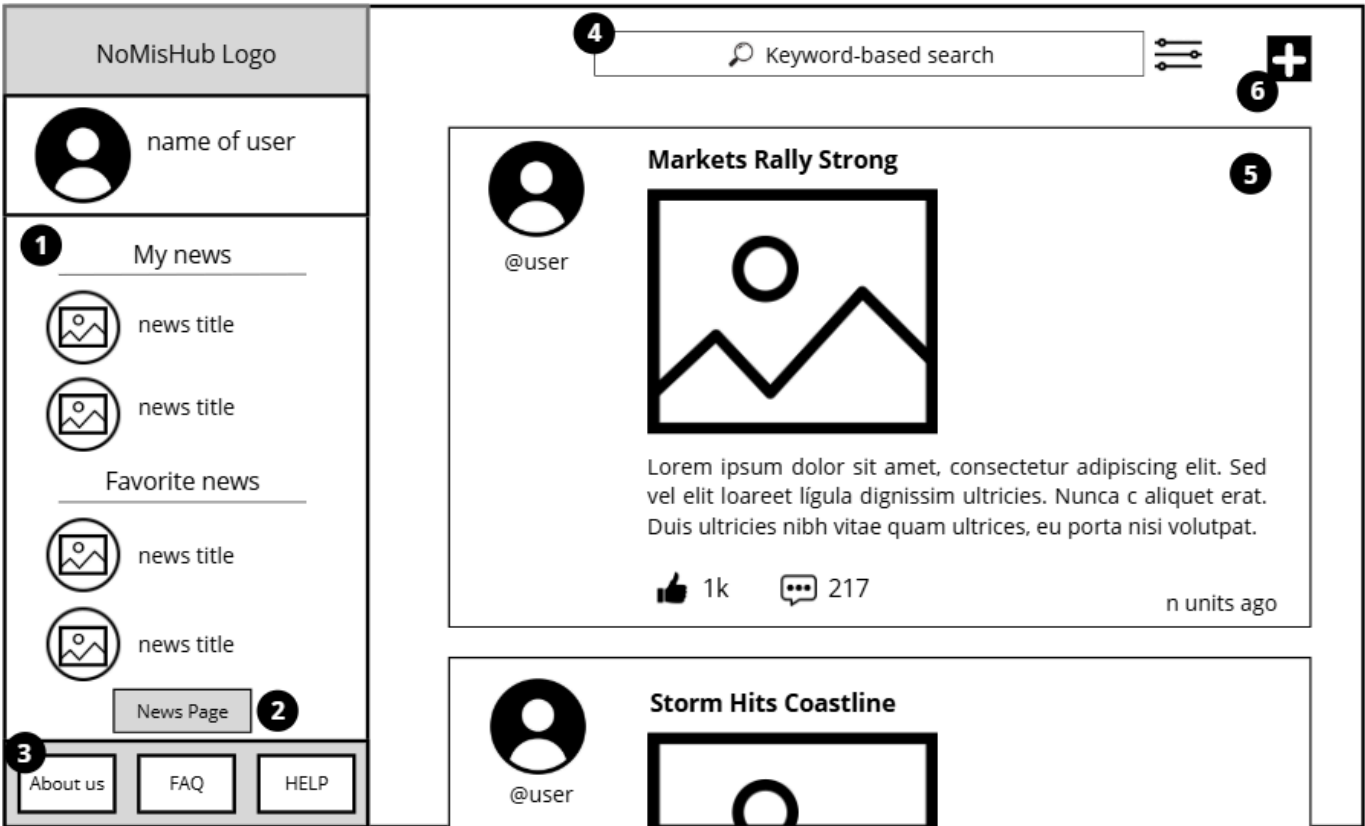


Figure 3: Home Page Wireframe.

1. Side-bar for quick access to news
2. Button to reach the News Page
3. Buttons that open pages with help and developer information.
4. Direct access to search feature
5. Area to interact with the latest news.
6. Post button.

UI02: News Page

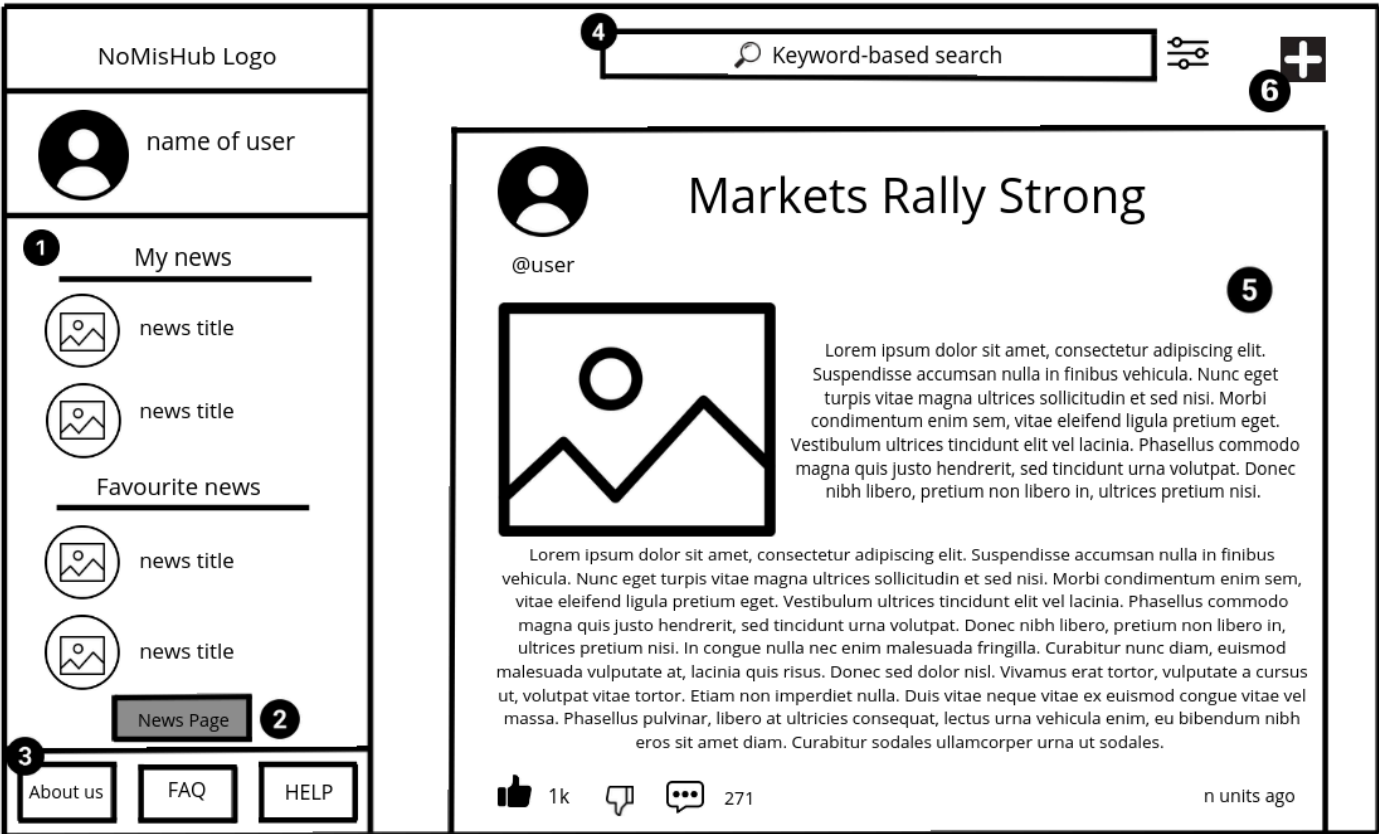


Figure 4: Full News Page Wireframe.

1. Side-bar for quick access to news
2. Button to reach the News Page
3. Buttons that open pages with help and developer information.
4. Direct access to search feature
5. Area to interact with the full news.
6. Post button.

UI03: Comments Section Page.

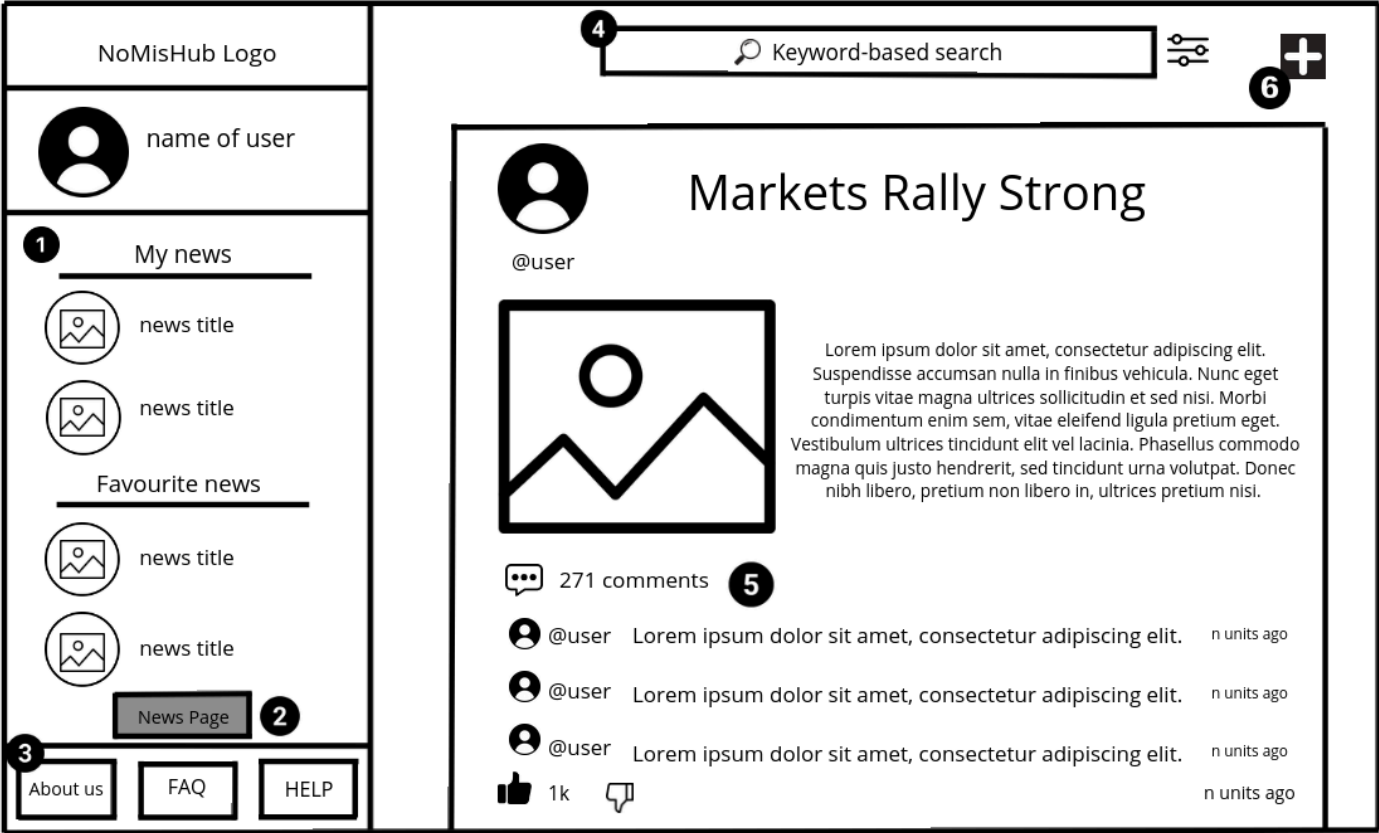


Figure 5: Comments Section Page Wireframe.

- 1. Side-bar for quick access to news
- 2. Button to reach the News Page
- 3. Buttons that open pages with help and developer information.
- 4. Direct access to search feature
- 5. Area to interact with the comments section.
- 6. Post button.

Revision history

Changes made to the first submission:

- 1. Updated Editor.
- 2. Merged overlapping US30 and US31.
- 3. Updated sitemap to have HomePage on the center.

GROUP2511, 04/10/2025

- Ana Catarina Barbosa Patrício, up202107383@fe.up.pt
- António Pedro Alves Pais, up202305444@fe.up.pt
- Hugo Ribeiro Alves, up202305395@fe.up.pt
- Tomás Pereira da Silva, up202307796@fe.up.pt

Editor: Hugo Ribeiro Alves

Comments